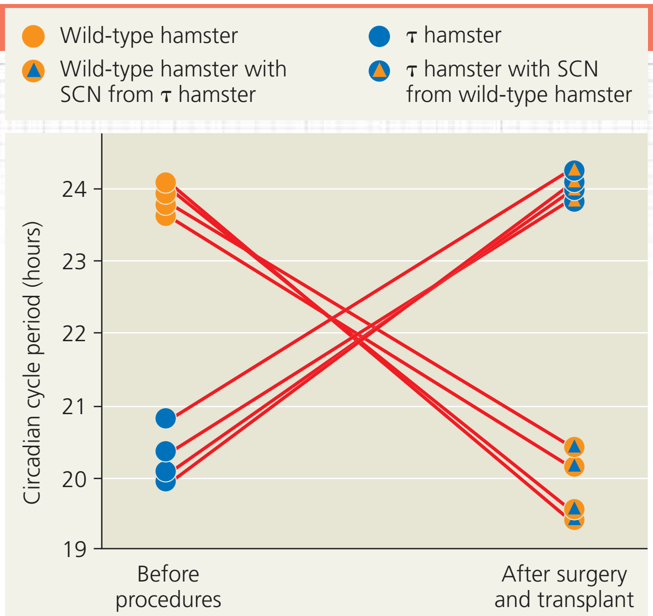
Scientific Skills Exercise

Designing an Experiment Using Genetic Mutants

Does the SCN Control the Circadian Rhythm in Hamsters? By surgically removing the SCN from laboratory mammals, scientists demonstrated that the SCN is required for circadian rhythms. Those experiments did not, however, reveal whether circadian rhythms originate in the SCN. To answer this question, researchers performed an SCN transplant experiment on wild-type and mutant hamsters (*Mesocricetus auratus*). Whereas wild-type hamsters have a circadian cycle lasting about 24 hours in the absence of external cues, hamsters that are homozygous for the τ (tau) mutation have a cycle lasting only about 20 hours. In this exercise, you will evaluate the design of this experiment and propose additional experiments to gain further insight.

How the Experiment Was Done The researchers surgically removed the SCN from wild-type and τ hamsters. Several weeks later, each of these hamsters received a transplant of an SCN from a hamster of the opposite genotype. To determine the periodicity of rhythmic activity for the hamsters before the surgery and after the transplants, the researchers measured activity levels over a three-week period. They plotted the data collected for each day in the manner shown in Figure 40.9a and then calculated the circadian cycle period.

Data from the Experiment In 80% of the hamsters in which the SCN had been removed, transplanting an SCN from another hamster restored rhythmic activity. For hamsters in which an SCN transplant restored a circadian rhythm, the net effect of the two procedures (SCN removal and replacement) on the circadian cycle period is graphed at the upper right. Each red line connects the two data points for an individual hamster.



Data from M. R. Ralph et al., Transplanted suprachiasmatic nucleus determines circadian period, *Science* 247:975–978 (1990).

INTERPRET THE DATA

1. In a controlled experiment, researchers manipulate one variable at a time. **(a)** What was the variable manipulated in this study? **(b)** Why did the researchers use more than one hamster for each procedure? **(c)** What traits of the individual hamsters would likely have been held constant among the treatment groups?
2. For the wild-type hamsters that received τ SCN transplants, what would have been an appropriate experimental control?
3. **(a)** What general trends does the graph above reveal about the circadian cycle period of the transplant recipients? **(b)** Do the trends differ for the wild-type and τ recipients? Based on these data, what can you conclude about the role of the SCN in determining the period of the circadian rhythm?
4. **(a)** In 20% of the hamsters, there was no restoration of rhythmic activity following the SCN transplant. What are some possible reasons for this finding? **(b)** How confident are you in your conclusion about the role of the SCN based on data from 80% of the hamsters?
5. Suppose that researchers identified a mutant hamster that lacked rhythmic activity; that is, its circadian activity cycle had no regular pattern. Propose SCN transplant experiments using such a mutant along with **(a)** wild-type and **(b)** τ hamsters. Predict the results of those experiments in light of your conclusion in question 3(b).

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

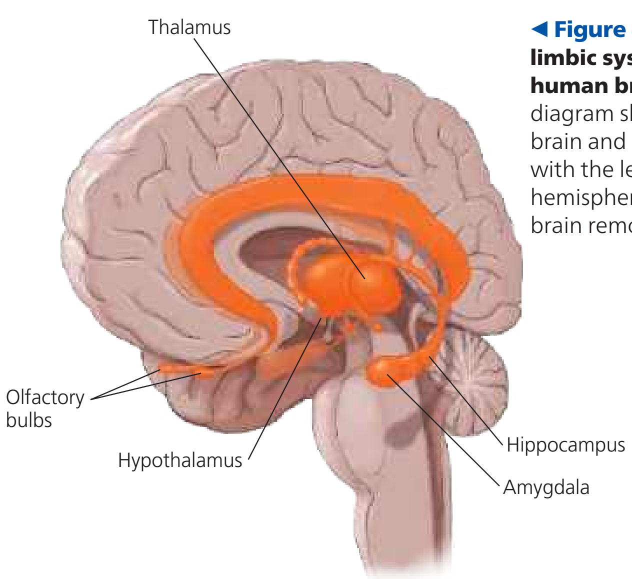
data from an experiment and propose experiments to test the role of the SCN in hamster circadian rhythms.

Mastering Biology HHMI Video: The Human Suprachiasmatic Nucleus

Emotions

Whereas a single structure in the brain controls the biological clock, the generation and experience of emotions depend on many brain structures, including the amygdala, hippocampus, and parts of the thalamus. As shown in **Figure 49.14**, these structures border the brainstem in mammals and are therefore called the *limbic system* (from the Latin *limbus*, border).

One way the limbic system contributes to our emotions is by storing emotional experiences as memories that can be recalled by similar circumstances. This is why, for example, a situation that causes you to remember a frightening event



◀ **Figure 49.14 The limbic system of the human brain.** This diagram shows the brain and brainstem, with the left cerebral hemisphere of the brain removed.